Automated License Plate Recognition for Parking Permit Enforcement at Cal Poly Pomona

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Problem

Cal Poly Pomona enforces its paid parking permits very inefficiently, leading many students to avoid paying for a permit to save costs. **We need a better way to enforce parking permits to free up more space for students who pay for permits.**

Background

Finding parking at Cal Poly Pomona is a persistent problem for students. Limited parking availability can cause students to arrive late to class and adds unnecessary stress to their daily commute. **At the same time, parking enforcement is still performed manually, requiring staff to check permits and issue citations by hand**. This inefficient enforcement process increases illegal parking of vehicles, **contributing to parking lot congestion and reducing availability for students who have purchased permits. In addition, the school loses potential revenue, as limited enforcement allows drivers to avoid parking fees.**

CPP Parking Fees [3]	
Semester Rate (vehicles)	$231.00
Semester Rate (motorcycles)	$92.00
Daily Parking Rate	$8.00
No Valid Permit/ No Permit Displayed (citation)	$80.00

Figure 1: Current Parking Rates & Citation Fines at CPP

Solution

To address the challenges of unauthorized parking and lost revenue, we propose the implementation of an **Automated License Plate Recognition (ALPR)** system. The ALPR camera will be installed at the entrance of CPP parking lots and will scan plates of cars entering as they pass by. By using high resolution cameras that feed into image recognition programs, the system will be able to reference vehicle license plates against the University's permit database and more efficiently enforce parking with a permit. In doing this, the number of citations a non-permit parker receives will get closer to and potentially beyond the 2.89 average per semester necessary to match the price of buying a parking permit, in effect making illegally parking fiscally unreasonable.

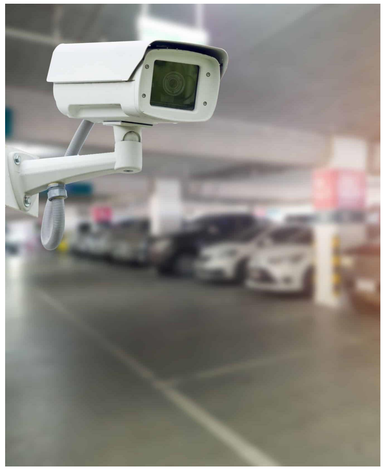

Figure 2: An example of an ALPR camera in a parking garage

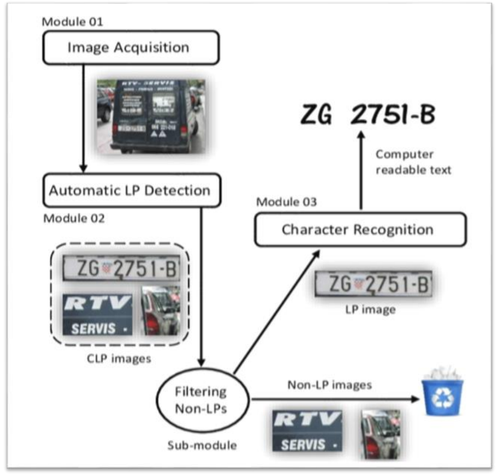

Figure 3: ALPR text extraction flowchart

Does this Work?

- **ALPR has proven to function at other institutions**, such as California State University, LA, Mt. SAC College, and U of W-La Crosse [1][4][6].
- **It's also more profitable than manual ticketing**: - "UWL's parking operations have seen a 27% increase in parking citations, and around 17% increase in permit revenue" [6]
- **Additionally, it allows universities to actually lower permit costs**: "We actually dropped the price of our permits this year because of LPR(License plate recognition),"[6]
- Although the implementation can be pricy, with Current predictions, **This system is expected to have a quick return on investment** for the University of Washington- La Crosse: "Hill believes UWL will see a full ROI in three to five years"[6]

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Pros and Cons

Advantages:

- ALPR parking permit enforcement will bring more money to CPP.
- More parking spots available to paying parkers
- Students moving to public transit and carpools to save costs, making more room in CPP lots while reducing carbon emissions.

Drawbacks:

- Implementation costs
- Digital Infrastructure for parking permits will need to be revised.

Conclusion

Based on the success of ALPR implementation at Cal State LA [1], Mt. San Antonio College [4], and University of Wisconsin-La Crosse [6], **bringing the technology to CPP would decrease parking violations** in our often-full lots. The extra revenue brought in by enforcing permits will allow CPP to **fund new parking solutions and lower parking rates, making permits affordable for the students**. In addition, students with permits will have an easier time finding spots with less parking violations.

References

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